

Generation Interconnection System Impact Study Report

For Queue Project Q19-1208

East Bismarck 100 kV

May 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1208. The Interconnection Customer proposes a 234.3 MW solar photovoltaic facility interconnecting at the East Bismarck 100 kV substation in the Illinois area. The request was received on May 18, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 464 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,136,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: MOUNTAIN ISLAND-TERMINOUS 100 kV
Contingency: TEXON-EAST BSMARCK 230 kV
Loading: 163.2% (Rating: 98 MVA)
DFAX: 37.8%
Pre-project loading: 72.3%
2. Facility: TERMINOUS-LAKE JACKSON 100 kV
Contingency: TEXON-E BISMARCK 230 kV
Loading: 147.2% (Rating: 128 MVA)
DFAX: 37.8%
Pre-project loading: 78.0%
3. Facility: LAKE JACKSON SUBSTATION-E BISMARCK 100 kV
Contingency: TEXON-EAST BISMARCK 230 kV
Loading: 109.1% (Rating: 131 MVA)
DFAX: 25.8%
Pre-project loading: 62.9%
4. Facility: CRUM CANYON SUBSTATION-N CHARLOTTE 100 kV
Contingency: TEXON-EAST BISMARCK 230 kV
Loading: 103.5% (Rating: 139 MVA)
DFAX: 19.5%
Pre-project loading: 70.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MOUNTAIN ISLND-TERMINOUS 100 kV	Rebuild line with higher-capacity conductor	\$3,273,000	72% (\$2,374,000)
TERMINOUS-LAKE JACKSON 100 kV Line	Rebuild line with higher-capacity conductor	\$2,278,000	72% (\$1,652,000)
LAKE JACKSON-EAST BISMARCK 100 kV line	Reconductor line	\$797,000	75% (\$597,000)
N CHARLOTTE-CRUM CANYON 100 kV	Reconductor line	\$4,998,000	30% (\$1,513,000)
	Total allocated to project		\$6,136,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.